

Data Share

Project 1: quantifying cell movement from microscopy data

dataset 1.1: cells moving with no drug added

migration_no_drug_low_density.pkl, migration_no_drug_med_density.pkl, migration_no_drug_high_density.pkl

Cell centroids are saved in `A_list`, which has shape (t, x, y). For each time point t, there is a spatial field of cells in the xy-plane, where 1's denote location of cells. See metadata for spatial length scale and interval between time points.

```
# example of loading and plotting data:  
library(tweenr)
```

```
## Warning: package 'tweenr' was built under R version 4.2.3
```

```
output <- py_load_object("migration_no_drug_med_density.pkl")
```

```
A_list <- output$A_list  
metadata <- output$metadata
```

```
cat("Size of data (t, 2D position): ", paste(dim(A_list), collapse = " x "), "\n")
```

```
## Size of data (t, 2D position):
```

```
cat("\nData Details:\n")
```

```
##
```

```
## Data Details:
```

```
for (key in names(metadata)) {  
  cat(key, "-", as.character(metadata[[key]]), "\n")  
}
```

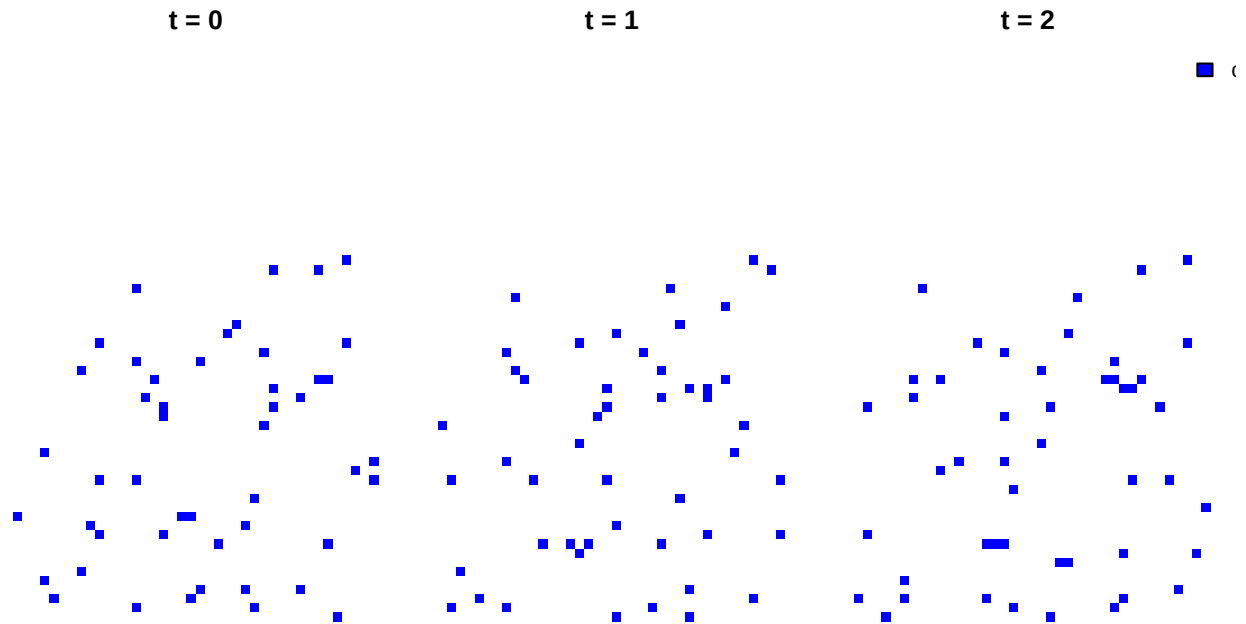
```
## experiment - no drug  
## cell density - med  
## movie frames - 41  
## xy dimensions - 40  
## microns per lattice space - 1  
## time interval (mins) - 5
```

```

# discrete colormap: 0 = empty (white), 1 = cell (blue)
cmap <- c("white", "blue")

par(mfrow = c(1, 3), mar = c(1, 1, 2, 1))
for (i in 1:3) {
  plot_frame(extract_frame(A_list, i), cols = cmap, main = paste("t =", i - 1))
}
legend("topright", inset = c(-0.2, 0), legend = "cells", fill = "blue",
      xpd = TRUE, bty = "n")

```



dataset 1.2: cells moving with drug added migration_with_drug_low_density.pkl, migration_with_drug_med_density.pkl, migration_with_drug_high_density.pkl

Cell centroids are saved in `A_list`, which has shape (t, x, y) . For each time point t , there is a spatial field of cells in the xy -plane, where 1's denote location of cells. See metadata for spatial length scale and interval between time points.

```

# load and plot like above

```

Project 2: determining effect of drug on cell-to-cell virus transmission

dataset 2.1: virus spreading with no drug added `infection_no_drug_experiment_0.pkl`, `infection_no_drug_experiment_1.pkl`, `infection_no_drug_experiment_2.pkl`

Cell centroids are again saved in `A_list`, which has shape `(t, x, y)`. For each time point `t`, there is a spatial field of cells in the `xy`-plane, where each value encodes the cell's state: 0 = no cell, 1 = susceptible cell, 2 = infected cell, 3 = recovered cell. For example, `A_list[t, x, y] == 2` means the cell at position `(x, y)` is infected at frame `t`.

See metadata for spatial length scale and interval between time points.

The biologists collected 3 movies (experiments 0, 1, and 2).

```
# example of loading and plotting data:
```

```
output <- py_load_object("infection_no_drug_experiment_0.pkl")
```

```
A_list <- output$A_list
```

```
metadata <- output$metadata
```

```
cat("Size of data (t, 2D position): ", paste(dim(A_list), collapse = " x "), "\n")
```

```
## Size of data (t, 2D position):
```

```
cat("\nData Details:\n")
```

```
##
```

```
## Data Details:
```

```
for (key in names(metadata)) {  
  cat(key, "-", as.character(metadata[[key]]), "\n")  
}
```

```
## experiment - infection without drug
```

```
## movie frames - 401
```

```
## xy dimensions - 40
```

```
## microns per lattice space - 1
```

```
## time interval (mins) - 0.5
```

```
# discrete colormap: one color per state (0=empty, 1=S, 2=I, 3=R)
```

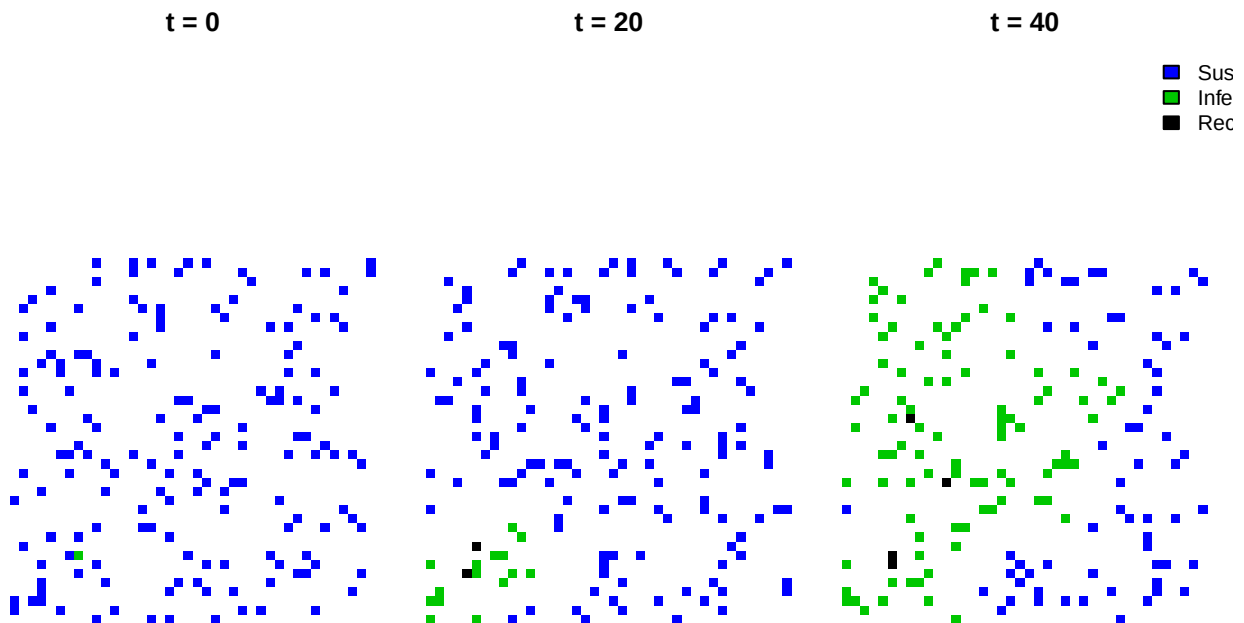
```
cmap <- c(  
  "white",           # 0 empty  
  "blue",           # 1 susceptible  
  rgb(0, 200 / 256, 0), # 2 infected  
  "black"           # 3 recovered  
)
```

```
times_to_plot <- c(0, 20, 40) # 0-indexed, as in the Python version
```

```

par(mfrow = c(1, length(times_to_plot)), mar = c(1, 1, 2, 1))
for (t in times_to_plot) {
  plot_frame(extract_frame(A_list, t + 1), cols = cmap, main = paste("t =", t))
}
legend("topright", inset = c(-0.3, 0),
      legend = c("Susceptible", "Infected", "Recovered"),
      fill = cmap[2:4], xpd = TRUE, bty = "n")

```



dataset 2.2: virus spreading with drug added `infection_with_drug_experiment_0.pkl` `infection_with_drug_experiment_1.pkl` `infection_with_drug_experiment_2.pkl`

Cell centroids are again saved in `A_list`, which has shape (t, x, y) . For each time point t , there is a spatial field of cells in the xy -plane, where each value encodes the cell's state: 0 = no cell, 1 = susceptible cell, 2 = infected cell, 3 = recovered cell. For example, `A_list[t, x, y] == 2` means the cell at position (x, y) is infected at frame t .

See metadata for spatial length scale and interval between time points.

The biologists collected 3 movies (experiments 0, 1, and 2).

```
# load and plot like above
```